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12

HEALTH AND STRESS

Are you a victim of "techno-stress"? In his classic book on the subject, Craig Brod (1984) defined *technostress* as "a modern disease of adaptation caused by an inability to cope with the new computer technologies in a healthy manner. It manifests itself in two distinct ways: in the struggle to accept computer technology, and in the more specialized form of over-identification with computer technology" (p. 16). When some of the technologies we now take for granted were first introduced, these potentially time- and labour-saving devices implied the possibility of a better future, with more time for leisure and social relationships. Instead, our world has simply adapted to the new technology by requiring that all aspects of our lives, and especially our work or school life, operate at an increasingly fast pace.

In the past few years it has become apparent that some people—especially the young—are not only undisturbed by the speed of communication but in fact prefer rapid and short bursts of communication such as texting or Twitter. However, others often feel overwhelmed by the rapid pace of change in technological gadgetry and the need for the constant contact it represents. In the last few years, Larry Rosen, past chair and professor of psychology at California State University, has provided much research evidence to support the view that our dependence on technology is affecting us negatively. In his most recent book (Rosen, Cheever, & Carrier, 2012), Rosen argues that our overreliance on technology fosters "obsessions, dependence and stress reactions" that can cause significant psychological problems or, as he calls it, an "idisorder." Just a few years ago, Canadian research showed that even information technology (IT) professionals feel quite overwhelmed by the pressure to keep on top of technologies that change so rapidly (Tsai, Compeau, & Haggerty, 2007). Today the Information Age is increasingly being blamed for making simple things more complicated—and stressing us out in the process.

As you will learn in this chapter, stress—any type of stress—affects our ability to function and tends to have a negative impact on our health. That we should never take our health for granted is good advice indeed, but it's advice that most of us easily forget. In this chapter we discuss many issues related to health and stress.

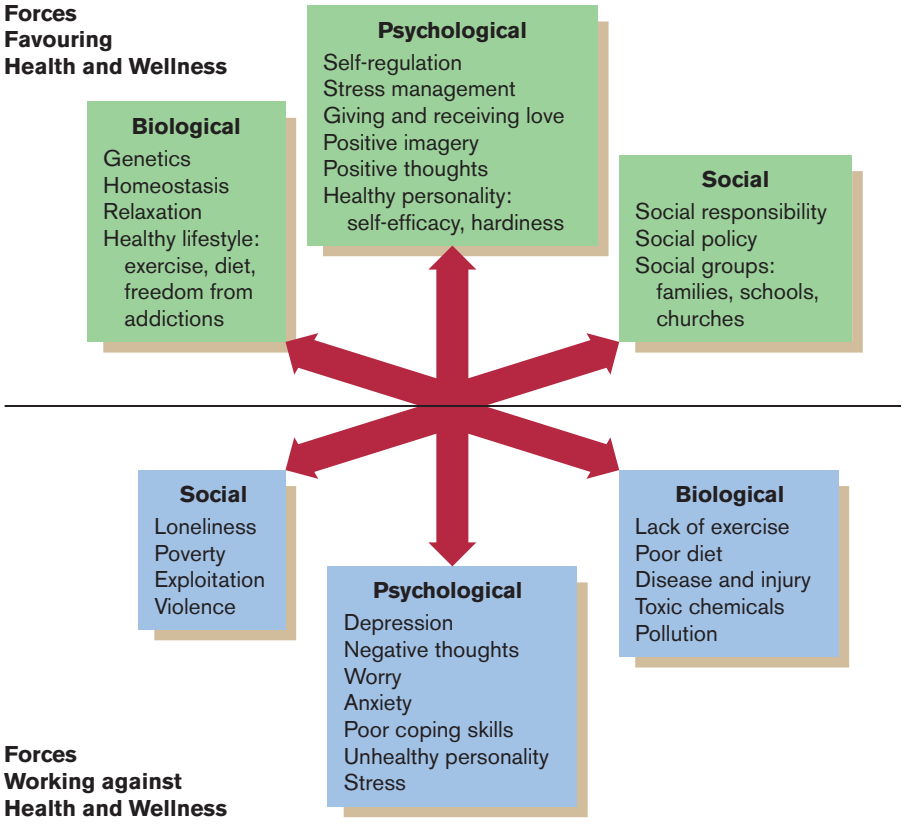


FIGURE 12.1
The Biopsychosocial Model of Health and Wellness
The biopsychosocial model focuses on health as well as illness, and holds that both are determined by a combination of biological, psychological, and social factors. Most health psychologists endorse the biopsychosocial model.
(From J. A. Green & Shellenberger, 1990.)

There are two main approaches to health and illness. The **biomedical model**, the predominant view in medicine, focuses on illness rather than health. It explains illness in terms of biological factors without considering psychological and social factors that might contribute to the condition.

Another approach that is gaining serious attention is the **biopsychosocial model** of health and wellness (see Figure 12.1). This approach focuses on health as well as illness, and holds that both are determined by a combination of biological, psychological, and social factors (Engel, 1977, 1980; G. E. Schwartz, 1982). Growing acceptance of this model gave rise to the subfield of **health psychology**, "the field within psychology devoted to understanding psychological influences on how people stay healthy, why they become ill, and how they respond when they do get ill" (S. E. Taylor, 1991, p. 6). Health psychology is particularly important today because several prevalent diseases, including heart disease and cancer, are related to unhealthy lifestyles and stress (S. E. Taylor, 2008).

Why do people become ill in the modern age? At the beginning of the twentieth century, the primary causes of death in Canada were pneumonia and infectious diseases such as diphtheria and tuberculosis. The health menaces of modern times are diseases related to unhealthy lifestyle and stress—heart attack, stroke, hardening of the arteries, cancer, and cirrhosis of the liver. In this chapter we will discuss stress, disease, and behaviours that promote and compromise health.

MODULE 12A THEORIES OF STRESS

LO 12.1 Describe Selye's concept of general adaptation syndrome.
LO 12.2 Describe the physiological responses during the alarm, resistance, and exhaustion stages of general adaptation.
LO 12.3 Explain the four phases of response to a potentially stressful event according to Lazarus's cognitive theory of stress and coping.
LO 12.4 Describe the primary and secondary appraisal processes that occur in response to a potentially stressful event.

How would you define *stress*? Is stress something in the environment? Is it a physiological or psychological reaction that occurs within a person? Most psychologists define **stress** as the physiological and psychological response to a condition that threatens or challenges the individual and requires some form of adaptation or adjustment.

HANS SELYE AND THE GENERAL ADAPTATION SYNDROME

An early, classic contribution to stress research was made by Walter Cannon (1932), who described the fight-or-flight response. Cannon discovered that when any threat is perceived by an organism (animal or human), the sympathetic nervous system and the endocrine glands prepare the body to



Hans Selye

fight the threat or flee from it. Cannon considered the fight-or-flight response wonderfully adaptive, because it helps the organism respond rapidly to threats. He also considered it potentially harmful in the long run if an organism is not able to fight or flee and experiences prolonged stress and continuing physical arousal (Sapolsky, 1994).

Canadian scientist Hans Selye (1907–1982) is the researcher most prominently associated with the study of stress and health. Selye spent most of his pioneering career at McGill University (1932–1945) and the Université de Montréal (1945–1977). At McGill, he conducted research on the effects of sex hormones. In one experiment he injected rats with hormone-rich extracts of cow ovaries. What happened to the rats? To Selye's amazement, (1) their adrenal glands became swollen, (2) their immune systems were weakened, and (3) they developed bleeding ulcers in the stomach and intestines. Never before had a hormone been shown to cause such clear physical symptoms. Selye thought he might be on the verge of discovering a new hormone. But after further experiments he found that he could produce the same symptoms by trying almost anything on the rats—for example, exposing them to toxic chemicals or freezing temperatures. Even extreme muscle fatigue caused the same symptoms. It seemed that Selye had not discovered anything at all. He was crushed. Then brooding gave way to reflection.

He realized that the body responds in much the same way to all harmful agents (toxic substances, injuries, electric shock) and a host of other stressors. The physical response was so predictable, so general, that Selye named it the **general adaptation syndrome (GAS)**. As a medical student in the 1920s, Selye had been struck by the fact that patients admitted to the hospital with an amazingly wide variety of illnesses all had many of the same physical symptoms. Now he was seeing general symptoms in rats exposed to a variety of stressors.

Selye was elated by his discovery, but the medical world was skeptical. The notion that organisms react in the same way to a wide range of dangers was completely contrary to the orthodox medical thinking of the day. Within five years, however, Selye had proved that the general stress reaction was indeed the body's way of responding to stress.

The General Adaptation Syndrome: A General Physical Response to Many Stressors

What is the general adaptation syndrome?

Selye knew that all living organisms are constantly confronted with **stressors**—stimuli or events that place a demand on the organism for adaptation or readjustment. Each stressor causes both specific and non-specific responses. Extreme cold, for example, causes the *specific* response of shivering. Apart from this, the body makes a *non-specific* response to a wide variety of stressors. The heart of Selye's concept of stress is the *general adaptation syndrome*, his term for the non-specific response to stress (see Figure 12.2, on the next page). The GAS consists of three stages: alarm, resistance, and exhaustion (Selye, 1956). [LO 12.1]

The body's first response to a stressor is the **alarm stage**, when emotional arousal occurs and the body prepares its defensive forces to meet the threat. [LO 12.2] In the alarm stage the sympathetic nervous system, through the release of hormones, mobilizes the body to fight or flee. If the stressor cannot be quickly conquered or avoided, the organism enters the **resistance stage**, which is characterized by intense physiological efforts to either resist or adapt to the stressor. [LO 12.2] During the resistance stage, the adrenal glands pour out powerful hormones (glucocorticoids) to help the body resist stressors.

biomedical model: A perspective that focuses on illness rather than health, explaining illness in terms of biological factors without regard to psychological and social factors.

biopsychosocial model: A perspective that focuses on health as well as illness, and holds that both are determined by a combination of biological, psychological, and social factors.

health psychology: The field concerned with the psychological factors that contribute to health, illness, and response to illness.

stress: The physiological and psychological response to a condition that threatens or challenges a person and requires some form of adaptation or adjustment.

general adaptation syndrome (GAS): The predictable sequence of reactions (the alarm, resistance, and exhaustion stages) that organisms show in response to stressors.

stressors: Any events capable of producing physical or emotional stress.

alarm stage: The first stage of the general adaptation syndrome, when there is emotional arousal and the defensive forces of the body are prepared for fight or flight.

resistance stage: The second stage of the general adaptation syndrome, during which there are intense physiological efforts to resist or adapt to the stressor.

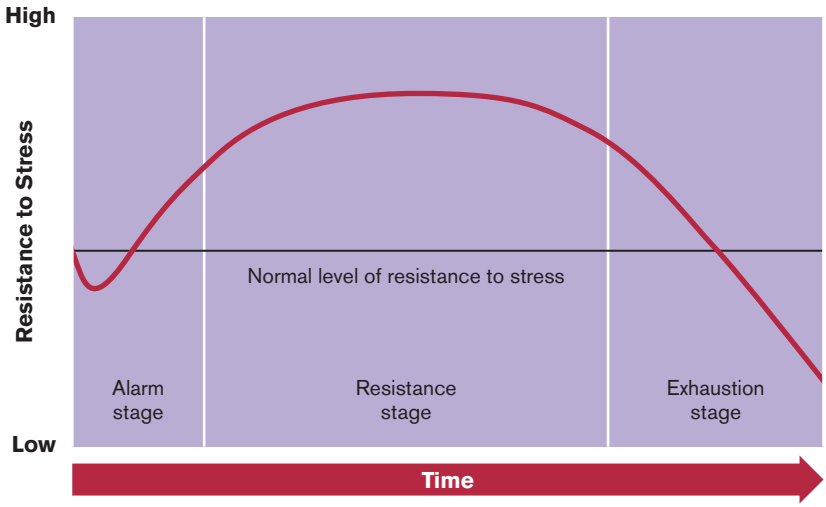


FIGURE 12.2
The General Adaptation Syndrome
The three stages in Hans Selye's general adaptation syndrome are (1) the alarm stage, during which there is emotional arousal and the defensive forces of the body are mobilized for fight or flight; (2) the resistance stage, in which intense physiological efforts are exerted to resist or adapt to the stressor; and (3) the exhaustion stage, when the organism fails in its efforts to resist the stressor.
(Based on Selye, 1956.)

Resistance may last a long time. According to Selye (1956), the length of the resistance stage depends both on the strength or intensity of the stressor and on the body's power to adapt.

If the organism fails in its efforts to resist, it reaches the **exhaustion stage**: "The stage of exhaustion after a temporary demand upon the body is reversible, but the complete exhaustion of all stores of deep adaptation energy is not" (Selye, 1974, p. 29). [LO 12.2] If exposure to the stressor continues, all the stores of deep energy are depleted, and disintegration, disease, or death may follow.

Selye claimed that any event requiring a readjustment, positive or negative, will produce stress in an organism. He did, however, differentiate between the positive and negative aspects of stress. *Eustress* is positive or good stress, including exhilaration, excitement, and the thrill of accomplishment. *Distress* is damaging or unpleasant stress, such as frustration, inadequacy, loss, disappointment, insecurity, helplessness, or desperation.

Criticisms of Selye's Theory: A Missing Cognitive Factor

The connection between extreme, prolonged stress and certain diseases is now widely accepted by medical experts, but some criticism of Selye's work seems justified. The major criticism is directed at Selye's contention that the intensity of the stressor determines one's physical reaction to it. His theory does not provide for a psychological component—that is, it does not consider how a person perceives and evaluates the stressor. This criticism led to the development of the cognitive theory of stress.

exhaustion stage: The final stage of the general adaptation syndrome, occurring if the organism fails in its efforts to resist the stressor.

primary appraisal: Evaluating the significance of a potentially stressful event according to how it will affect one's well-being–whether it is perceived as irrelevant or as involving harm or loss, threat, or challenge.

secondary appraisal: Evaluating one's coping resources and deciding how to deal with a stressful event.

RICHARD LAZARUS'S COGNITIVE THEORY OF STRESS

Richard Lazarus (1922–2002) argued that it is not the stressor itself that causes stress, but a person's perception of the stressor (Lazarus, 1966; Lazarus & Folkman, 1984). Because Lazarus emphasized the importance of perceptions and the appraisal of stressors, his is a *cognitive* theory of stress and coping. To Lazarus, the stress process can be understood in terms of four phases. [LO 12.3] First, there is a causal agent, either external or internal, commonly referred to as *stress* or *the stressor*. Second, the mind or the body evaluates the stressor as either threatening or benign. Third, the mind or the body uses coping processes to deal with the stressor. Finally, there is the stress reaction—the "complex pattern of effects on mind and body" (Lazarus, 1993, p. 4). Lazarus believed that physiological and psychological stress must be analyzed differently. He argued that whereas Selye's general adaptation syndrome describes how the body copes with physiological stress, his model focuses on how we cope with psychological stressors.

The Cognitive Appraisal of Stressors: Evaluating the Stressor and Considering Your Options

What are the roles of primary and secondary appraisal when people are confronted with a potentially stressful event?

According to Lazarus, when people are confronted with a potentially stressful event, they engage in a cognitive process that involves a primary and a secondary appraisal. A **primary appraisal** is an evaluation of the meaning and significance of a situation—whether its effect on our well-being is positive, irrelevant, or negative. [LO 12.4] An event appraised as negative or stressful could involve (1) harm or loss—damage that has already occurred; (2) threat—the potential for harm or loss; or (3) challenge—the opportunity to grow or gain. An appraisal of threat, harm, or loss can be made in relation to anything important to us—a friendship, a part of our body, our property, our finances, or our self-esteem.

The same event can be appraised differently by different people. Some students may welcome the opportunity to give an oral presentation in class, seeing it as a challenge and a chance to impress their professor and raise their grade. Other students may feel threatened, fearing that they may embarrass themselves in front of their classmates and lower their grade in the process. Still others may view the assignment as both a challenge and a threat. When we appraise a situation as one involving harm, loss, or threat, we have negative emotions such as anxiety, fear, anger, and resentment (Folkman, 1984). A challenge appraisal, on the other hand, is usually accompanied by positive emotions such as excitement, hopefulness, and eagerness. When we assess an event as stressful, we engage in a **secondary appraisal**. During secondary appraisal, if we judge the situation to be within our control, we make an evaluation of our available coping resources: physical (health, energy, stamina), social (support network), psychological (skills, morale, self-esteem), material (money, tools, equipment), and time. **[LO 12.4]** Then we consider our options and decide how we will deal with the stressor. The level of stress we feel depends largely on whether our resources are adequate to cope with the threat, and how severely our resources will be taxed in the process. Figure 12.3 summarizes Lazarus and Folkman's psychological model of stress.

Research generally supports Lazarus and Folkman's contention that the physiological, emotional, and behavioural reactions to stressors depend partly on whether the stressors are appraised as challenging or threatening. Recent Canadian research also suggests that some stressful situations can even be anticipated in advance, allowing people to use a strategy called *proactive coping*, which consists of efforts or actions taken in advance of a potentially stressful situation to prevent its occurrence or to minimize its consequences (Greenglass & Fiksenbaum, 2009). Proactive copers anticipate and then prepare for upcoming stressful events and situations.

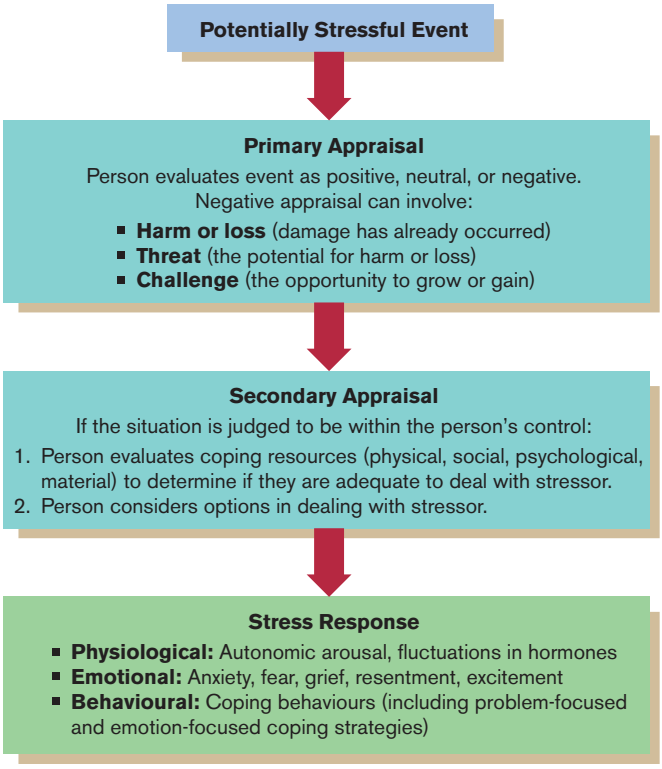


FIGURE 12.3
Lazarus and Folkman's Psychological Model of Stress
Lazarus and Folkman emphasize the importance of a person's perceptions and appraisal of stressors. The stress response depends on the outcome of the primary and secondary appraisals, whether the person's coping resources are adequate to cope with the threat, and how severely the resources are taxed in the process.
(Based on Folkman, 1984.)

REMEMBER IT

Theories of Stress

1. Selye focused on the _____aspects of stress; Lazarus focused on the _____aspects of stress.
   a. physiological; physiological
   b. physiological; psychological
   c. psychological; physiological
   d. psychological; psychological
2. Which stage of the general adaptation syndrome is marked by intense physiological efforts to adapt to the stressor?
   a. the readjustment stage
   b. the resistance stage
   c. the alarm stage
   d. the exhaustion stage
3. Susceptibility to illness increases during what stage of the general adaptation syndrome?
   a. the readjustment stage
   b. the resistance stage
   c. the alarm stage
   d. the exhaustion stage
4. During secondary appraisal, we
   a. evaluate our coping resources and consider options in dealing with the stressor.
   b. determine whether an event is positive, neutral, or negative.
   c. determine whether an event involves loss, threat, or challenge.
   d. determine whether an event causes physiological or psychological stress.

Answers: 1. b 2. b 3. d 4. a

MODULE 12B SOURCES OF STRESS: THE COMMON AND THE EXTREME

LO 12.5 Describe approach–approach, approach–avoidance, and avoidance–avoidance conflicts in motivation.
LO 12.6 Explain how the unpredictability of and lack of control over a stressor affect its impact.
LO 12.7 Explain the relationship between racism and the experience of stress.
LO 12.8 Identify the stages in which victims tend to react to catastrophic events.
LO 12.9 Describe the symptoms that characterize post-traumatic stress disorder.

Some stressors produce temporary stress, whereas others produce chronic stress—a state of stress that continues unrelieved over time. Chronic health problems, physical handicaps, poverty, and unemployment are sources of chronic stress. The burden of chronic stress is disproportionately heavy for the poor, for minorities, and for the elderly.

EVERYDAY SOURCES OF STRESS

How do approach–approach, avoidance–avoidance, and approach–avoidance conflicts differ?

Sometimes conflicting motives can be sources of stress. When we must make a choice between desirable alternatives, we are facing an **approach–approach conflict**, and stress may be the result. [LO 12.5] Some approach–approach conflicts are minor, such as deciding which movie to see. Others can have major consequences, such as whether to continue building a promising career or to interrupt the career to raise a child.

In an **avoidance–avoidance conflict**, we must choose between two undesirable alternatives. [LO 12.5] You may want to avoid studying for an exam but at the same time want to avoid failing the test. In an **approach–avoidance conflict**, we are simultaneously drawn to and repelled by a choice; for example, we may want to take a wonderful vacation but we would have to empty our savings account to do so. [LO 12.5]

Unpredictability and Lack of Control: Factors That Increase Stress

How do the unpredictability of and lack of control over a stressor affect its impact?

Our physical and psychological well-being is profoundly influenced by the degree to which we feel a sense of control over our lives (Rodin & Salovey, 1989). [LO 12.6] Ellen J. Langer and Judith Rodin (1976) studied the effects of control on nursing-home residents. One group of residents was given some measure of control over their lives, such as choices in arranging their rooms and in the times they could see movies. They showed improved health and well-being and had a lower death rate than another group that was not given control. Within 18 months, 30 percent of the residents given no choices had died, compared with only 15 percent of those who had been given some control over their lives.

Several studies suggest that we are less subject to stress when we believe we have control over the source of stress, whether we exercise that power or not. It is likely that our expectations about unpredictable stressors may influence the manner in which we respond to them. For instance, in one study, experimenters exposed participants to an annoying noise (a randomly sounded buzzer) but offered them the opportunity to control the intensity of the noise (Bollini, Walker, Hamann, & Kestler, 2004). Even though all participants had an equal opportunity to exert some control over the stressor, only those who had a general belief in their own ability to moderate the effects of the stressor appeared to benefit. These participants were found to have lower levels of the stress hormone cortisol than participants who believed they lacked the ability to control stressors.

Racism and Stress

A significant source of chronic stress is being a member of a minority group in a majority culture (Ong, Fuller-Rowell, & Burrow, 2009). A study of white and black participants' responses to a survey about ways of coping suggests that a person may experience racial stress from simply being one of the few members of a particular race in any of a variety of settings (such as a classroom, workplace, or social setting). [LO 12.7] The feeling of stress can be intense, even in the absence of racist attitudes, discrimination, or any other overt evidence of racism (Plummer & Slane, 1996).

Some theorists have proposed that *historical racism*—the history of repression of members of particular groups—can also be a source of stress. Historical racism has been studied most extensively with African Americans in the United States, and many of these researchers who focus on this line of research claim that the higher incidence of high blood pressure and other chronic medical conditions among African Americans is attributable to stress associated with historical racism (D. R. Williams & Mohammed, 2009). Surveys have shown that members of that group report experiencing more race-related stress than do members of other minority groups (Utsey, Chae, Brown, & Kelly, 2002). Researchers now argue that this phenomenon must be studied more thoroughly with other historically repressed groups, such as Aboriginal peoples (Belcourt-Dittloff & Stewart, 2000).

CATASTROPHIC EVENTS AND CHRONIC INTENSE STRESS

How do people typically react to catastrophic events?

Do you remember how you felt when you found out about the terrorist attacks taking place in the United States on September 11, 2001, or when you first heard of the earthquake in Haiti in 2010 or the tsunami that devastated Japan in 2011? Like many people, you were probably a bit stunned and emotionally numb. Although the reactions of those who are directly affected are usually more extreme, catastrophes are stressful not only for people who experience them directly but also for those who learn of the events via news media.

CANADIAN CONNECTIONS

War and Post-traumatic Stress Disorder

In the spring of 2003, as Canadian soldiers were preparing to increase their participation in the peacekeeping efforts in Afghanistan, former Canadian Forces ombud André Marin warned the military that "Post Traumatic Stress Disorder is an operational hazard that is a fact of modern peacekeeping missions." In his report, Marin noted that nearly 20 percent of those in uniform and 50 percent of reservists experience PTSD (Dineen, 2003). He also noted that soldiers suffering from PTSD-related illnesses were often stigmatized by their peers and superiors, and that many were considered "fakers." Regrettably, a follow-up report suggests that little has changed since the release of the ombud's original report ("Post Traumatic," 2009).

Since the beginning of Canada's military involvement in Afghanistan, roughly 25 percent of Canadian soldiers have returned from this mission with mental health problems. The most recent assessment suggests that 12 percent of soldiers had one or more mental-health problems, including post-traumatic stress disorder (PTSD) and depression. Another 13 percent of them reported "harmful" or "hazardous" drinking. Canadian Forces ombudsman Pierre Daigle has called PTSD and related stress illnesses "a real hardship for Canada's soldiers, sailors, airmen and airwomen for many years to come" (Sher, 2011). ■

Victims tend to react to catastrophic events in a series of stages. [LO 12.8] First, they seem disoriented and may wander about aimlessly, often unaware of their own injuries and without attempting to help themselves or others. Following this stage, victims show a concern for others; although unable to act efficiently on their own, they are willing to follow the directions of rescue workers. As victims begin to recover, this shock is replaced by generalized anxiety.

Recovering victims typically have recurring nightmares and feel a compulsive need to retell the event over and over. Perhaps re-experiencing the event through dreaming and retelling helps desensitize them to the horror of the experience. Crisis intervention therapy can provide victims with both coping strategies and realistic expectations about the problems they may face in connection with the trauma. Although most people eventually manage the stress associated with such catastrophes quite well, some people experience an extreme form of stress reaction known as *post-traumatic stress disorder*.



When writer Pico Iyer's home was destroyed by arson, he saved only his cat and his manuscript. A fairly common reaction to such catastrophic events is post-traumatic stress disorder.

POST-TRAUMATIC STRESS DISORDER

What is post-traumatic stress disorder?

Post-traumatic stress disorder (PTSD) is a prolonged and severe stress reaction to a catastrophic event (such as a plane crash, an earthquake, or rape) or to chronic intense stress, such as that which occurs in combat or during imprisonment as a hostage. PTSD can also be the result of living in a neighbourhood in which violent crime is a daily occurrence (Kilpatrick et al., 2003) or even in an environment wherein one is exposed to domestic violence (Griffin et al., 2006). The disorder may show up immediately, or it may not appear until six months or more after the traumatic experience, in which case it is called *delayed* post-traumatic stress disorder. The most serious cases of PTSD are found among those who have witnessed brutal atrocities: victims of sexual violence and child abuse (Filipas & Ullman, 2006; Hedtke et al., 2008), Holocaust survivors (Kuch & Cox, 1992), and soldiers who have experienced combat (M. Price, Gros, Strachan, Ruggiero, & Acierno, 2012). The *Canadian Connections* box looks at the PTSD suffered by soldiers returning from combat.

approach–approach conflict: A conflict arising from having to choose between desirable alternatives.

avoidance–avoidance conflict: A conflict arising from having to choose between equally undesirable alternatives.

approach–avoidance conflict: A conflict arising when a choice has both desirable and undesirable features, so that you are both drawn to and repelled by the same choice.

post-traumatic stress disorder (PTSD): A prolonged and severe stress reaction to a catastrophic or otherwise traumatic event; characterized by anxiety, psychic numbing, withdrawal from others, and the feeling that one is reliving the traumatic experience.

REMEMBER IT

Sources of Stress

1. Rick cannot decide whether to go out or stay home and study for his test. What kind of conflict does he have?
   a. approach–approach conflict
   b. avoidance–avoidance conflict
   c. approach–avoidance conflict
   d. ambivalence–ambivalence conflict
2. Victims of catastrophic events typically want to talk about their experience. (true/false)
3. What has research shown to increase stress?
   a. predictability of the stressor
   b. unpredictability of the stressor
   c. predictability of and control over the stressor
   d. unpredictability of and lack of control over the stressor
4. Post-traumatic stress disorder is a prolonged and severe stress reaction that results when a number of common sources of stress occur simultaneously. (true/false)

Answers: 1. c 2. true 3. d 4. false

People with post-traumatic stress disorder often have flashbacks, nightmares, or intrusive memories in which they feel as if they are actually re-experiencing the traumatic event. [LO 12.9] They suffer increased anxiety and startle easily, particularly in response to anything that reminds them of the trauma (American Psychiatric Association, 2000). Many survivors of war or catastrophic events experience *survivor guilt* because they lived while others died; some feel that perhaps they could have done more to save others (Southwick, Gilmartin, Mcdonough, & Morrissey, 2006). People who develop PTSD are also quite likely to experience depression (Bleich, Koslowsky, Dolev, & Lerer, 1997) and cognitive difficulties such as poor concentration (Vasterling et al., 2002).

MODULE 12c COPING WITH STRESS

LO 12.10 Describe coping.
LO 12.11 Explain and compare problem-focused and emotion-focused coping.
LO 12.12 Describe an effective stress management strategy.

When we encounter stressful situations, we try either to alter them or to reinterpret them to make them seem more favourable. **Coping** refers to our efforts to deal with demands that we perceive as taxing or overwhelming (Lazarus, 1993). [LO 12.10]

PROBLEM-FOCUSED AND EMOTION-FOCUSED COPING

What is the difference between problem-focused and emotion-focused coping?

Coping strategies fall into two categories: problem focused and emotion focused (Lazarus & Folkman, 1984). **Problem-focused coping** is direct; it involves reducing, modifying, or eliminating the source of stress. [LO 12.11] If you are getting a poor grade in history and appraise this as a threat, you may study harder, talk over your problem with your professor, form a study group with other class members, get a tutor, or drop the course.

But what can we do when we face stress that we cannot fight, escape from, avoid, or modify in any way? We can

REVIEW & REFLECT 12.1 PROBLEM-FOCUSED AND EMOTION-FOCUSED COPING STRATEGIES

Coping Strategy	Definition	Examples
Problem focused	A response aimed at reducing, modifying, or eliminating the source of stress.	Acting to remove or lessen the threat. Removing oneself from the stressful situation. Enlisting the help of others in dealing with the threat. Seeking professional help or advice. Acting to prevent recurrence of similar stressful situations.
Emotion focused	A response aimed at reducing the emotional distress caused by the stressor.	Viewing the stressor as a challenge rather than a threat. Using one of these responses: prayer, denial, wishful thinking, fantasizing, humour, relaxation, biofeedback, alcohol, drugs, overeating, promiscuous sex.

REMEMBER IT

Coping with Stress

1. Lazarus's research focused on the _____ aspects of coping.
   a. complex
   b. psychological
   c. physiological
   d. emotion-focused
2. Experts who have studied how to cope with traumatic events suggest that it is often best to isolate yourself from others in order to cope successfully. (true/false)
3. Coping aimed at reducing, modifying, or eliminating a source of stress is called _____ coping; coping aimed at reducing an emotional reaction to stress is called _____ coping.
   a. emotion-focused; problem-focused
   b. problem-focused; emotion-focused
   c. primary; secondary
   d. secondary; primary
4. Researchers recommend that people use a combination of problem-focused and emotion-focused coping when dealing with a stressful situation. (true/false)

Answers: 1. b 2. false 3. b 4. true

use **emotion-focused coping** to change the way we respond emotionally. [LO 12.11] Emotion-focused coping may involve reappraising a stressor by using many possible strategies. Some people may use prayer, denial, or wishful thinking. They may also fantasize about various options or use humour. Others may prefer to use relaxation, biofeedback, or even alcohol, drugs, overeating, or promiscuous sex in order to reduce the emotional impact of a serious problem. A person who loses his job, for example, may decide that it isn't a major tragedy and instead view it as a challenge—an opportunity to find a better job with a higher salary. And despite what you may have heard, ignoring a stressor, one form of emotion-focused coping, can be an effective way of managing stress. For example, in a study of 116 people who had experienced heart attacks, all the participants reported being worried about suffering another attack. However, those who tried to ignore their worries were less likely to exhibit anxiety-related symptoms such as nightmares and flashbacks (Ginzburg, Solomon, & Bleich, 2002). Other emotion-focused strategies, such as keeping a journal in which you write about your worries and track how they change over time, may be even more effective (Alford, Malouff, & Osland, 2005). But a combination of problem-focused and emotion-focused coping is probably the most effective stress management strategy (Folkman & Lazarus, 1980). [LO 12.12] *Review & Reflect 12.1* summarizes the problem-focused and emotion-focused coping strategies.

MODULE 12D EVALUATING LIFE STRESS: MAJOR LIFE CHANGES, HASSLES, AND UPLIFTS

LO 12.13 Describe what the Social Readjustment Rating Scale is designed to measure.

LO 12.14 Explain the connection between life stress and health problems.

LO 12.15 Explain the balance between daily hassles and uplifts in the experience of stress.

There are two major approaches to evaluating life stress and its relation to illness. One approach focuses on major life events, which cause life changes that require adaptation. A second approach focuses on life's daily hassles.

HOLMES AND RAHE'S SOCIAL READJUSTMENT RATING SCALE: ADDING UP THE STRESS SCORES

What is the Social Readjustment Rating Scale designed to reveal?

Interested in the relationship between life changes and illness, Thomas Holmes and Richard Rahe (1967) developed the **Social Readjustment Rating Scale (SRRS)**. The SRRS is designed to measure stress by ranking different life events from most to least stressful. [LO 12.13] Each life event is assigned a point value. Life events that produce the greatest life changes and require the greatest adaptation are considered the most stressful, regardless of whether the events are positive or negative. The original scale assesses whether a person has experienced one of 43 life events that include the death of a spouse (100 stress points) to such items as divorce (73 points), death of a close family member (63 points), marriage (50 points), pregnancy (40 points), and trouble with the boss (23 points) to minor law violations such as getting a traffic ticket (11 points). More recent research has shown that scores

coping: Efforts through action and thought to deal with demands that are perceived as taxing or overwhelming.

problem-focused coping: A response aimed at reducing, modifying, or eliminating a source of stress.

emotion-focused coping: A response aimed at reducing the emotional impact of the stressor.

Social Readjustment Rating Scale (SRRS): A stress scale, developed by Holmes and Rahe, which ranks 43 life events from most to least stressful and assigns a point value to each.

on the scale developed by Holmes and Rahe in 1967 are still correlated with a variety of health indicators (De Coteau, Hope, & Anderson, 2003; Dohrenwend, 2006; Thorsteinsson & Brown, 2009).

Holmes and Rahe maintain that there is a connection between the degree of life stress and major health problems. [LO 12.14] After analyzing more than 5000 medical case histories, they concluded that major life changes often precede serious illness (Rahe, Meyer, Smith, Kjaer, & Holmes, 1964). However, one of the main shortcomings of the SRRS is that it assigns a point value to each life change without taking into account whether the change is for the better or the worse. For example, life changes such as divorce, separation, pregnancy, retirement from work, and changing jobs or residences may be either welcome or unwelcome.



Even positive life events, such as getting married, can cause stress.

Paul Insel and Walter Roth (1985) developed a similar scale to assess the degree of stress experienced by students. Would you like to learn the number of stress points in your life? Complete *Try It*.

THE HASSLES OF LIFE: LITTLE THINGS STRESS A LOT

What roles do hassles and uplifts play in the stress of life, according to Lazarus?

Which do you think is more stressful—major life events or those little problems and frustrations that seem to crop up every day? Richard Lazarus believes that the little stressors, which he calls **hassles**, add up to more stress than major life events. Daily hassles are the "irritating, frustrating, distressing demands and troubled relationships that plague us day in and day out" (Lazarus & DeLongis, 1983, p. 247). [LO 12.15] Lazarus and colleagues

TABLE 12.1 The 11 Most Common Hassles for College and University Students

Hassle
1. Not having enough time for class assignments and studying, balancing social relationships, and organizing one's schedule
2. Not enough money for housing, school, entertainment, or clothes
3. Racial and ethnicity-based conflicts including feeling treated differently because of race, perception of others based on cultural stereotypes, or feeling discriminated against
4. Not being taken seriously or denied opportunities because of gender
5. Friendship concerns including being lonely, not having enough friends, or difficulty making friends
6. Traffic and parking problems
7. Feeling unaccepted or mocked due to religion
8. Safety for person or fear of having material stolen
9. Problems with current job including lack of satisfaction or scheduling
10. Physical appearance including weight concerns and getting into shape
11. Parental expectations including demanding parents and lack of independence

Source: Pett, M. A., & Johnson, M. J. M. (2005). Development and psychometric evaluation of the revised University student hassle scale. *Educational and Psychological Measurement,* 65(6), 984–1010. Copyright © 2005, SAGE Publications. Used with permission.

TRY IT

Student Stress Scale

The Student Stress Scale is scored by adding the points listed for the checked life events. Insel and Roth suggest the following interpretations: Scores of 300 and higher indicate a relatively high health risk; scores of 150 to 299 indicate a 50–50 chance of serious health problems within two years. What stress management skills that you have studied could you apply to the results of this scale?

Instructions: Check those events you have experienced in the past six months or are likely to experience in the next six months. Add up the points to derive your total stress score.

Rank	Life Event	Past	Future	Stress Value	Your Scores
1	Death of a close family member	___	___	100	___
2	Death of a close friend	___	___	73	___
3	Divorce between parents	___	___	65	___
4	Jail term	___	___	63	___
5	Major personal injury or illness	___	___	63	___
6	Marriage	___	___	58	___
7	Fired from job	___	___	50	___
8	Failed important course	___	___	47	___
9	Change in health of a family member	___	___	45	___
10	Pregnancy	___	___	45	___
11	Sex problems	___	___	44	___
12	Serious argument with a close friend	___	___	40	___
13	Change in financial status	___	___	39	___
14	Change of major	___	___	39	___
15	Trouble with parents	___	___	39	___
16	New girl or boy friend	___	___	38	___
17	Increased workload	___	___	37	___
18	Outstanding personal achievement	___	___	36	___
19	First quarter/semester in college	___	___	35	___
20	Change in living conditions	___	___	31	___
21	Serious argument with instructor	___	___	30	___
22	Lower grades than expected	___	___	29	___
23	Change in sleeping habits	___	___	29	___
24	Change in social activities	___	___	29	___
25	Change in eating habits	___	___	28	___
26	Chronic car trouble	___	___	26	___
27	Change in number of family get-togethers	___	___	26	___
28	Too many missed classes	___	___	25	___
29	Change of college	___	___	24	___
30	Dropped more than one class	___	___	23	___
31	Minor traffic violations	___	___	20	___
				Total:	___

Source: Insel & Roth, 1985.

(Kanner, Coyne, Schaefer, & Lazarus, 1981) developed the Hassles Scale to assess various categories of hassles include irritating, frustrating experiences such as standing in line, being stuck in traffic, waiting for an appliance or utility repair technician to come to your home, and so on. People completing the scale indicate which items have been a hassle for them and rate them for severity on a three-point scale. Research indicates that minor hassles that accompany stressful major life events are better predictors of the level of psychological distress than are the major life events themselves (Pillow, Zautra, & Sandler, 1996). **[LO 12.15]**

According to Lazarus, "A person's morale, social functioning, and health don't hinge on hassles alone, but on a balance between the good things that happen to people—that make them feel good—and the bad" (quoted in Goleman, 1979, p. 52). Table 12.1 shows the 11 most frequent hassles reported by students.

hassles: Little stressors that include the irritating demands and troubled relationships that are encountered daily and that, according to Lazarus, cause more stress than do major life changes.

REMEMBER IT

Evaluating Life Stress

1. On the Social Readjustment Rating Scale, only negative life changes are considered stressful. (true/false)
2. The Social Readjustment Rating Scale takes account of the individual's perceptions of the stressfulness of the life change in assigning stress points. (true/false)
3. According to Lazarus, hassles typically account for more life stress than do major life changes. (true/false)
4. Lazarus's approach in measuring hassles and uplifts considers individual perceptions of stressful events. (true/false)

Answers: 1. false 2. false 3. true 4. true

Fortunately, life's **uplifts**—that is, the positive experiences—may neutralize or cancel out many of the hassles. [LO 12.15] Lazarus and his colleagues also constructed an Uplifts Scale. As with the Hassles Scale, people make a cognitive appraisal in determining what they consider an uplift. Items viewed as uplifts by some people may actually be stressors for other people. For example, middle-aged people, uplifts are often health or family related (Pinquart & Sörensen, 2004), whereas for college students uplifts often take the form of having a good time (Kanner et al., 1981).

MODULE 12E HEALTH AND DISEASE

LO 12.16 Identify positive strategies for coping with cancer and cancer treatment.
LO 12.17 Describe the psychological impact of HIV infection and AIDS and strategies for coping psychologically.
LO 12.18 Describe psychoneuroimmunology and the effects of stress and depression on the immune system.
LO 12.19 Explain the role of optimism in physical health.
LO 12.20 Discuss social support as a factor that contributes to better health.

Health psychologists study the myriad ways in which we respond to illness risk factors, and the factors that affect whether we seek treatment. Let's look at two examples: cancer and AIDS.

CANCER: A DREADED DISEASE

Cancer. The word alone is frightening. Cancer is second only to heart disease as the leading cause of death. The Canadian Cancer Society estimated in 2011 that in that year, 177 800 Canadians would be diagnosed with cancer and 75 000 would die of it (Canadian Cancer Society, 2011). Young people are not spared the scourge of cancer, which takes the lives of more children between the ages of 3 and 14 than any other disease.

We speak of cancer as a single disease, but actually it is a complicated collection of diseases. Cancer can invade the cells in any part of a living organism—humans, other animals, and even plants. Cancer always starts small, because it is a disease of the body's cells. Normal cells in all parts of the body reproduce (divide), and they have built-in instructions about when to stop doing so. If they did not, every part of the body would continue to grow as long as it lived. Unlike normal cells, cancerous cells do not stop dividing. Unless they can be caught in time and destroyed, they continue to grow and spread, eventually killing the organism.

Risk Factors for Cancer

Health psychologists warn that an unhealthy diet, smoking, excessive alcohol consumption, promiscuous sexual behaviour, and becoming sexually active in the early teens (especially for females) are all behaviours that increase the risk of cancer. Compared with those who do not get cancer, many cancer patients report that they faced more high-stress situations in their lives before their cancer was diagnosed.

Coping with Cancer

What can cancer patients do to help themselves cope with having cancer?

People who are diagnosed with cancer must adjust to the chronic stressors associated with it. They must cope with difficult medical therapies along with "continued emotional distress, disrupted life tasks, social and interpersonal turmoil and fatigue and low energy" (B. L. Anderson, Kiecolt-Glaser, & Glaser, 1994, p. 390). The chronic stress associated with cancer can damage the autonomic, endocrine, and immune systems. Researchers suggest that patients need more than medical treatment: their therapy should also involve helping them maintain their quality of life. Patients should be able to discuss their fears and anxieties, be given information about their disease and treatment, and be taught how to lower their arousal. [LO 12.16] In addition, research conducted at the University of Manitoba (Hack, Degner, Watson, & Sinha, 2006) has shown that women who exercise some control over their treatment "had significantly higher physical and social functioning and significantly less fatigue than women who assumed a passive role" (p. 9). [LO 12.16]

What have health psychologists found that can help cancer patients? Carver and colleagues (1993, 2005) found that breast-cancer patients who maintained an optimistic outlook, accepted the reality of their situation, and maintained a sense of humour experienced less distress three months and six months after surgery. Patients who refused to accept the reality of the situation

and who had thoughts of giving up experienced much higher levels of distress. Dunkel-Schetter, Feinstein, Taylor, and Falke (1992) found that the most effective elements of a strategy for coping with cancer were social support (such as through self-help groups), a focus on the positive, and distraction. [LO 12.16] Avoidant coping strategies such as fantasizing, denial, and social withdrawal were associated with more emotional distress.

AIDS

What happens to a person from the time of infection with HIV to the development of full-blown AIDS?

The most feared disease related to the immune system is AIDS (acquired immune deficiency syndrome), which is caused by the human immunodeficiency virus (HIV). The virus attacks the helper cells, gradually but relentlessly weakening the immune system. The first case of AIDS was diagnosed in Canada in the early 1980s; there is still no cure for it and no vaccine to protect against it. Finding a cure is urgent—worldwide statistics for 2010 estimate that 34 million people around the world live with HIV or AIDS (Avert.org, 2011).

When a person is first infected, HIV enters the bloodstream. This initial infection usually causes no symptoms, and the immune system begins to produce HIV antibodies. It is these antibodies that are detected in the AIDS test. Individuals then progress to the asymptomatic carrier state, during which they experience no symptoms at all and thus can unknowingly infect others.

HIV attacks the immune system until it becomes essentially non-functional. The diagnosis of AIDS is made when the immune system is so damaged that victims develop rare forms of cancer or pneumonia or other "opportunistic" infections. Such infections would not usually affect people with a normal immune response; in people who have a very impaired immune system, these infections can be life-threatening. At this point, patients typically experience progressive weight loss, weakness, fever, swollen lymph nodes, and diarrhea; 25 percent develop a rare cancer that produces red-purple spots on the skin. Other infections develop as the immune system weakens further.

The Transmission of AIDS

HIV is transmitted primarily through the exchange of blood, semen, or vaginal secretions during sexual contact or when IV (intravenous) drug users share contaminated needles or syringes (Des Jarlais & Friedman, 1994). Infected mothers can also infect the fetus prenatally, during childbirth, and when breastfeeding. The most effective treatment to date is a combination of drugs that include one type known as protease inhibitors (Collier et al., 1996), but this treatment must be carefully monitored (Liu et al., 2006).

The Psychological Impact of HIV Infection and AIDS

Most people are psychologically devastated when they are diagnosed with the AIDS virus. [LO 12.17] Not only are they being sentenced to an early death, but there is a social stigma associated with AIDS that few other diseases have.

To cope psychologically, AIDS patients and those infected with HIV need education and information about the disease. [LO 12.17] They can be helped by psychotherapy, self-help groups, and medications such as antidepressants and anti-anxiety drugs. Self-help groups and group therapy may serve as substitute family for some patients. An ever-present concern voiced by patients in psychotherapy is whether to tell others and, if so, what to tell them and how. Patients may feel a compelling need to confide in others and, at the same time, to conceal their condition.

Protection against Sexually Transmitted Diseases: Minimizing Risk

Although everyone has seen or heard the messages about "safer" behaviours, many people still ignore the message. Researchers in Ontario surveyed adults at dating bars (Herold & Mewhinney, 1993) and on spring break in Florida (Maticka-Tyndale, Herold, & Mewhinney, 1998) and found that even though people were concerned about contracting AIDS, many of them had not used any form of protection during their last sexual encounter. People who choose to practise promiscuous sex cannot be safe, but can reduce the risks by using a latex condom. Sexual abstinence or monogamous relationships with partners free of infection also greatly reduce risk.

STRESS AND THE IMMUNE SYSTEM

What are the effects of stress and depression on the immune system?

In a field of study known as **psychoneuroimmunology**, psychologists, biologists, and medical researchers combine their expertise to learn how psychological factors (emotions, thinking, and behaviour) affect the immune system. [LO 12.18]

Several studies show that emotions, psychological factors, and stress are related to immune system functioning (S. Cohen, 2004; Robles, Glaser, & Kiecolt-Glaser, 2005). [LO 12.18] These relationships are incredibly complex, given that our immune system is an interconnected defence system working with the brain to keep the body healthy (R. Ader, 2000). For example, social factors, such as close social ties to family, friends, and others, have been shown to have positive effects on the immune system, whereas poor marital relationships, sleep deprivation, exams, and academic pressures have been linked to lowered immune response (Kiecolt-Glaser et al., 1987; Kiecolt-Glaser, Glaser, Gravenstein, Malarkey, & Sheridan, 1996). This pattern of protection holds across age and racial groups, for both sexes, at all educational levels, and at every season of the year (R. Ader, 2000; S. Cohen, Doyle, Skoner, Rabin, & Gwaltney, 1997). Severe, incapacitating depression is related to lowered

uplifts: The positive experiences in life, which can neutralize the effects of many of the hassles.

psychoneuroimmunology (sye-ko-NEW-ro-IM-you-NOLL-oh-gee): A field in which psychologists, biologists, and medical researchers study the effects of psychological factors on the immune system.

immune activity (Robles, Shaffer, Malarkey, & Kiecolt-Glaser, 2006). After the death of a spouse, the widow or widower suffers weakened immune system function and is at a higher risk of mortality; this risk may last for up to two years following a partner's death (Prigerson et al., 1997).

Periods of high stress have been correlated with increased symptoms of many infectious diseases, including oral and genital herpes, mononucleosis, colds, and flu. Stress can cause decreased levels of the immune system's B and T cells. People who experience stress may indeed be more susceptible to coronary heart disease, stroke, and poorer pregnancy outcomes (N. Adler & Matthews, 1994; K. Davidson, MacGregor, Stuhr, Dixon, & MacLean, 2000). But even more ominous is the growing evidence that stress can impair the functioning of the immune system itself. [LO 12.18] For instance, people exposed to cold viruses are more likely to develop colds if they have experienced a greater number of life changes in the previous year (S. Cohen, Miller, & Rabin, 2001). In fact, physicians have long observed that stress and anxiety can worsen autoimmune diseases (Chrousos & Kino, 2007).

PERSONAL FACTORS REDUCING THE IMPACT OF STRESS AND ILLNESS

What three personal factors are associated with health and resistance to stress?

Researchers have identified three personal factors that may contribute to better health: optimism, psychological hardiness, and social support.

Optimism and Pessimism

Researchers have shown a strong relationship between positive attitudes and physical health (Pressman & Cohen, 2005). Recent longitudinal research has even shown that an optimistic outlook can reduce the risk of a first heart attack by 50 percent when compared to the least optimistic (Boehm, Peterson, Kivimaki, & Kubzandky, 2011). It should not be surprising, then, that people who are generally optimistic tend to cope more effectively with stress, and this in turn may reduce their risk of illness (Scheier, Carver, & Bridge, 2001; Seligman, 1990). [LO 12.19] Optimists generally expect good outcomes, and this helps make them more stress resistant than pessimists, who tend to expect bad outcomes. Researchers suspect that associations between optimism and resistance to stress are explained by the finding that, in response to similar stressors, optimists secrete lower levels of stress hormones than pessimists do (Lai et al., 2005). Optimism seems to be especially important in the prevention of stress-induced symptom flare-ups among patients with chronic disorders such as multiple sclerosis (Mitsonis, Potagas, Zervas, Sfagos, 2009).

Optimism is at the heart of a movement called *positive psychology*. The main proponent of the movement, Martin Seligman, argues that psychologists should focus more attention on using research to improve people's lives than they do on treating psychological disorders (Seligman & Csikszentmihalyi, 2000). Perhaps you too should consider taking a more positive outlook on life . . . doing so could have many social and health-related benefits.



A strong social support network can help a person recover faster from an illness.

Psychological Hardiness: Commitment, Challenge, and Control

Suzanne Kobasa (1979; Kobasa, Maddi, & Kahn, 1982) wondered why some people under great stress succumb to illness while others do not. She studied 670 male executives, who identified stressful life events and symptoms of illness that they had experienced in the preceding three years. She then administered personality questionnaires to 200 executives who ranked high for both stress and illness and to 126 who had faced equally stressful life events but had few symptoms of illness. She found three qualities that distinguished those who remained healthy from those who had a high incidence of illness: *commitment*, *control*, and *challenge*. Kobasa collectively called these psychological qualities **hardiness**.

Hardy individuals feel a strong sense of commitment to their work and personal life. They see themselves as having control over consequences and outcomes, and they welcome challenges. Being hardy also has some positive impact for students. Research has shown that hardiness can also reduce your academic stress and, consequently, reduce students' reported health complaints (Hystad, Eid, Laberg, Johnsen, & Bartone, 2009).

Social Support: Help in Time of Need

Another factor that seems to contribute to better health is **social support** (Thoits, 2011). [LO 12.20] This includes support provided by a spouse or other family members, friends, neighbours, colleagues, support groups, and members of the larger community. Social support can direct help, information, and advice to the individual, as well as emotional support. Social support provides the feeling that we are loved, valued, esteemed, and cared for by those to whom we feel a mutual obligation (S. Cohen, 1988, 2004).

Social support appears to have positive effects on the body's immune system as well as on the cardiovascular and endocrine systems (Uchino & Birmingham, 2011). Social support may encourage health-promoting behaviours and reduce the impact of stress so that people are less likely to resort to unhealthy methods of coping, such as smoking and drinking. Further, social support has been shown to reduce depression and enhance self-esteem in individuals who suffer from chronic

REMEMBER IT

Health and Disease

1. Lowered immune response has been associated with
   a. stress.
   b. depression.
   c. stress and depression.
   d. neither stress nor depression.
2. Some research suggests that optimists are more stress resistant than pessimists. (true/false)
3. Social support tends to reduce stress but is unrelated to health outcomes. (true/false)
4. Which of the following is not a dimension of psychological hardiness?
   a. a feeling that adverse circumstances can be controlled and changed
   b. a sense of commitment and deep involvement in personal goals
   c. a tendency to look upon change as a challenge rather than a threat
   d. close, supportive relationships with family and friends

Answers: 1. c 2. true 3. false 4. d

illnesses (Symister & Friend, 2003). People with social support recover more quickly from illnesses and lower their risk of death from specific diseases. Social support may even increase the probability of surviving a heart attack because it buffers the impact of stress on cardiovascular function (Steptoe, 2000).

Research on natural disasters reveals that initial social support is common but that the support tends to deteriorate because the needs of victims overwhelm the tangible and emotional resources of friends and family (Kaniasty & Norris, 2004). Other research has shown that high levels of perceived social support are associated with lower levels of depression (Sheets & Mohr, 2009). Such perceived support may be more a function of individual personality than of the actual availability of family and friends who can offer help. These results underscore the importance of psychological variables in health.

MODULE 12F YOUR LIFESTYLE AND YOUR HEALTH

LO 12.21 Explain factors that constitute an unhealthy lifestyle.
LO 12.22 Identify factors apart from the physical addiction that can diminish a smoker's ability to quit smoking.
LO 12.23 Describe the disease model of alcoholism and the limitations of this model.
LO 12.24 Identify the benefits of regular exercise to promoting better health.

What constitutes an unhealthy lifestyle, and how serious a factor is lifestyle in illness and disease?

If you are not healthy or physically hardy, who or what is to blame? There are a number of enemies of good health: environmental pollutants; job, family, and personal stressors; genetic and congenital defects; accidents and injury; and others. But the number one concern with most people is lifestyle habits. The specific culprits are all well known: an unhealthy diet, overeating, lack of exercise, alcohol and drug abuse, too much coffee, too little sleep, and so on. [LO 12.21] The most dangerous unhealthy behaviour of all is smoking.

SMOKING: HAZARDOUS TO YOUR HEALTH

Why is smoking considered the single most preventable cause of death?

According to the most recent Canadian statistics, 18 percent of Canadians aged 15 or older smoke (Health Canada, 2009). In 2011 alone, smoking was directly related to more than 37 000 deaths in Canada; "of those," Health Canada predicted at the time, "more than 300 non-smokers will die of lung cancer and at least 700 non-smokers will die of coronary heart disease caused by exposure to second-hand smoke" (Health Canada, 2011a). Add to these statistics the suffering from chronic bronchitis, emphysema, and other respiratory diseases; death and injury from fires caused by smoking; and low birth weight and impaired fetal development in babies born to mothers who smoke.

Why do adult smokers continue the habit even though most admit they would prefer to be non-smokers? There seems little doubt that smoking is an addiction. Nicotine is a powerful substance that increases the release of acetylcholine, norepinephrine, dopamine, and other neurotransmitters, which improve mental alertness, sharpen memory, and reduce tension and anxiety (Pomerleau & Pomerleau, 1989). Withdrawal symptoms from smoking typically last two to four weeks and the average smoker makes five or six attempts to quit before succeeding. But quitting smoking is more complex than overcoming the physical addiction. Rates of success vary substantially depending on the strategy used and the commitment made to quitting. [LO 12.22] For instance, research has shown that only 20 percent of smokers using nicotine replacement, such as patches or gum, succeed (Rose, 2006). Perceived stress

hardiness: A combination of three psychological qualities shared by people who can undergo high levels of stress yet remain healthy: commitment to one's personal goals, a sense of control over one's life, and a tendency to view change as a challenge rather than as a threat.

social support: Tangible support, information, advice, and/or emotional support provided in time of need by family, friends, and others; the feeling that we are loved, valued, and cared for.

may be an important contributor to successful quitting, with those who perceive that their lives are not very stressful having far greater success than those who feel more stress (S. B. Norman, Norman, Rossi, & Prochaska, 2006). Hence, planning a quitting date that corresponds with a period of lower stress may be a wise decision.

ALCOHOL: A PROBLEM FOR MILLIONS

What are some health risks of alcohol consumption?

The health and social costs of alcohol—in fatalities, lost work, family problems, and so on—are staggering. According to the most recent comprehensive study conducted in Canada, direct alcohol-related health care costs alone totalled $3.3 billion in 2002 (Rehm, Patra, & Popova, 2006). Alcohol abuse and dependence are significantly more prevalent among men than among women. People who begin drinking before age 15 are more likely to become dependent on alcohol than those who begin later (Grant & Dawson, 1998; Prescott & Kendler, 1999). Research suggests that alcohol is related to 6 percent of all deaths in Canada for people below the age of 70, with a much higher rate for men (7.6 percent) than women (3.5 percent) (Rehm, Patra, & Popova, 2006)

Alcohol can damage virtually every organ in the body, but it is especially harmful to the liver and is the major cause of cirrhosis. Alcoholics are about three times as likely to die in automobile accidents or of heart disease as non-alcoholics, and they have twice the rate of death from cancer. Pregnant women should avoid all alcohol because of its potentially disastrous effects on the developing fetus. Alcohol also affects the brain. For instance, research using MRI scans has shown damage to the brains of long-term alcoholics (Daurignac et al., 2005). Even shorter-term heavy drinking can cause cognitive damage and dysfunction that continues for several months after the drinking has stopped (Harper, 2009; E. V. Sullivan, Fama, Rosenbloom, & Pfefferbaum, 2002). The only good news in recent studies is that some of alcohol's effects on the brain seem to be partially reversible with prolonged abstinence.

Alcoholism's toll goes beyond physical damage to the alcoholic. Drunk drivers kill and injure. Alcohol has also been implicated in drownings, suicides, rapes, burglaries, and assaults.

Alcoholism: Causes and Treatment

The Canadian Medical Association maintains that alcoholism is a disease, and "once an alcoholic, always an alcoholic." According to this view, even a small amount of alcohol causes an irresistible craving for more, leading alcoholics to lose control of their drinking. Total abstinence is seen as the only acceptable recourse. The medical establishment and Alcoholics Anonymous endorse both the disease concept and the total abstinence approach to treatment.

Recent studies suggest that there is a genetic factor in alcoholism and lend support to the disease model. [LO 12.23] For example, neuroscience research suggests that the brains of alcoholics respond differently to visual and auditory stimuli than do those of non-alcoholics, and that these differences act as a genetic marker associated with a predisposition to alcoholism (Hada, Porjesz, Chorlian, Begleiter, & Polich, 2001; Porjesz et al., 2005; Prabhu et al., 2001). Relatives of alcoholics have been found to be more likely to become alcoholics themselves or to suffer from other types of addictions (Kamarajan et al., 2006; Zhang, Cohen, Projesz, & Begleiter, 2001).

The hypothesis that there is a greater genetic risk of alcoholism for men is also supported by years of research on identical and fraternal twins. [LO 12.23] Studies have revealed a substantial genetic influence (Kendler, Gardner, & Prescott, 2011; Tyndale, 2003). A study of 1000 pairs of female identical and fraternal twins found that alcoholism in women is 50 to 60 percent heritable, which is a rate similar to that for male alcoholics (Kendler, Gruenberg, & Kinney, 1994; Prescott & Kendler, 1999).

Is alcoholism a disease? Some experts reject the disease concept and contend that alcoholism can take various forms and have various causes. Some researchers caution against overlooking the environmental contribution to alcoholism, even in people who are genetically predisposed (Horwitz, Videon, Schmidtz, & Davis, 2003). [LO 12.23] Family and cultural influences have also been associated with problem drinking. For instance, a Canadian study (McCreary, Newcomb, & Sadava, 1999) demonstrated that men who endorsed traditional male gender roles engaged in a higher level of alcohol consumption than others.

EXERCISE: KEEPING FIT IS HEALTHY

What are some benefits of regular aerobic exercise?

For years medical experts, especially health psychologists, have promoted regular exercise. Yet only a small proportion of the population is highly active, and nearly 60 percent of the Canadian population aged 12 and older are physically inactive (Statistics Canada, 2002). Many studies show that regular **aerobic exercise** pays rich dividends in the form of physical and mental fitness. [LO 12.24] Aerobic exercise, such as running, swimming, brisk walking, bicycling, rowing, and jumping rope, is exercise that uses the large muscle groups in continuous, repetitive action and requires increased oxygen intake and increased breathing and heart rates. Aerobic exercise should be performed regularly—three or four times a week for 20 to 30 minutes, with additional 5- to 10-minute warm-up and cool-down periods (Centers for Disease Control and Prevention [CDC], 2006). The importance of regular, systematic aerobic exercise in keeping the cardiovascular system healthy cannot be overemphasized. This is true for people of all ages. Even preschoolers receive cardiovascular benefits from as little as an additional 60 minutes per week of exercise. This activity "may improve bone properties, aerobic fitness, and motor skills in some children" (Timmons, Naylor, & Pfeiffer, 2007). And regular, planned exercise yields dramatic increases in muscle and bone strength in older people. Strength training, for example, has been found to reduce sarcopenia, the age-related process in which muscles deteriorate. Such training also appears to prevent the loss of bone mass, called *osteoporosis* (CDC, 2006).

We do not need to become marathon runners to enjoy the maximum benefits of exercise, but individuals who engage in

aerobic exercise (eh-RO-bik): Exercise involving the use of large muscle groups in continuous, repetitive action and requiring increased oxygen intake and increased breathing and heart rates.



Strength training moderates the effects of aging on older adults' muscles and bones.

more than three hours of aerobic activity each week are more successful at losing excess weight and keeping it off than are those who exercise less (Votruba, Horvitz, & Schoeller, 2000). And a daily brisk walk of 30 minutes or more helps to reduce stress and yields the fitness standard associated with a much lower death rate.

In case you are not yet convinced, consider the following benefits of exercise and other small lifestyle changes that may improve your health:

Engage in moderate physical activity every day (e.g., walk up and down stairs for 15 minutes; spend 30 minutes washing a car). Such small amounts of activity will reduce your feelings of anxiety and sadness, will increase your bone density, and will reduce your risk of diabetes, heart disease, high blood pressure, and many other life-shortening diseases (Health Canada, 2011b). And it will also likely help you lose weight.

While we are on the topic of weight loss, if you are overweight, try to lose just 5 kilograms. Doing so will result in a 34 percent reduction in triglyceride levels, a 16 percent decrease in total cholesterol, and an 18 percent increase in HDL ("good" cholesterol); will significantly reduce your blood pressure; and will decrease your risk for diabetes, sleep apnea, and osteoarthritis (CDC, 2011).

Add just 20 to 30 grams of fibre to your diet each day. Doing so will improve your bowel function, reduce your risk of colon cancer and other digestive-system diseases, decrease your total blood cholesterol, reduce your blood pressure, and improve your insulin function (FiberFoods.net, 2012).

Of course, stop smoking! Doing so will result in immediate health improvement. It will improve your circulation, reduce your blood level of carbon monoxide, stabilize your pulse rate and blood pressure, improve your sense of smell and taste, improve your lung function and endurance, and reduce your risk of lung infections such as pneumonia and bronchitis. In the long term, stopping smoking will reduce your risk of lung cancer and decrease your likelihood of developing other smoking-related illnesses such as emphysema and heart disease (Health Canada, 2005).

Finally, consider limiting the amount of time you spend in front of the TV. Canadian psychologists Donald McCreary and Stanley Sadava, at Brock University (2000), looked at the relation between TV viewing and health, weight, and physical fitness. Women who spent a lot of time watching TV believed they were in poorer health. Overall, both men and women who watched a lot of TV thought they were overweight more than did people who watched less TV. And remember that spending endless hours on Facebook or on the Web will have similar negative effects on your health.

REMEMBER IT

Your Lifestyle and Your Health

1. Which is the most important factor leading to disease and death?
   - **a.** unhealthy lifestyle
   - **b.** a poor health care system
   - **c.** environmental hazards
   - **d.** genetic disorders
2. Which health-compromising behaviour is responsible for the most deaths?
   - **a.** overeating
   - **b.** smoking
   - **c.** lack of exercise
   - **d.** excessive alcohol use
3. (Alcohol/Smoking) damages virtually every organ in the body.
4. To improve cardiovascular fitness, aerobic exercise should be done
   - **a.** 15 minutes daily.
   - **b.** 1 hour daily.
   - **c.** 20 to 30 minutes daily.
   - **d.** 20 to 30 minutes 3 or 4 times a week

Answers: 1. a 2. b 3. Alcohol 4. d

APPLY IT

Interpreting Health Information on the Internet

All of us now use the internet to access information. From reviews of new music, to booking hotels for an upcoming trip, to answering any of our small queries, the Web has become a most valued tool. But how reliable is health information available on the internet? In an American Medical Association–sponsored large-scale study of health-related websites, researchers found that the quality of information varied widely from one site to another (Eysenbach, Powell, Kuss, & Sa, 2002). A study sponsored by the British Medical Association of internet-based advice for managing children's fever found that most websites contained erroneous information. Moreover, in a follow-up study done four years later, the researchers found that about half of the sites were no longer available; those that remained showed little improvement in the quality of information.

Despite these difficulties, physicians' organizations acknowledge the potential value of the internet in helping patients learn about and manage their own health. And because so many older adults are using the internet to learn about health issues, the AARP (formerly the American Association of Retired Persons) has published a list of points to keep in mind when surfing the Web for health information and advice (2002):

- Remember that there are no rules governing what is published on the internet. Unlike scientific journal articles, which are usually written and reviewed by experts in the field, internet articles can be posted by anyone, without review of any kind. Without expert knowledge, it is extremely difficult to tell whether the information and advice these articles contain are valid.
- Consider the source. Generally, Web sites sponsored by medical schools, government agencies, and public health organizations are reliable. Others, especially those promoting a health-related product, should be considered suspect.
- Get a second opinion. Ask your health care provider about internet-based information or read what's available from several different sources on the topic.
- Examine references. Sites that refer to credible sources (e.g., books, other Web sites) that you can find on the internet or in a library or bookstore are probably more reliable than sites that offer no references to support their advice.
- How current is the information? Health-related information changes frequently. Be certain that you are reading the most current findings and recommendations.
- Is it too good to be true? As in all areas of life, if something sounds too good to be true (e.g., a vitamin that cures cancer), it probably is. Try to find experimental, placebo-controlled studies that support any claims.

Using these guidelines, you can become a better consumer of internet-based health information.

THINKING CRITICALLY

Evaluation

Can people always cure themselves of illnesses? What are the limits to what people can do to help themselves?

Point/Counterpoint

Prepare two arguments, one supporting the position that alcoholism is a genetically inherited disease, and the other supporting the position that alcoholism is not a medical disease but results from learning.

Psychology in Your Life

Choose several stress-producing incidents from your own life and explain what problem-focused and emotion-focused coping strategies you used. From the knowledge you have gained in this chapter, list other coping strategies that might have been more effective.

MyPsychLab

GO TO MYPSYCHLAB (ACCESS CODE REQUIRED) TO FIND WEB RESOURCES FOR YOUR TEXT THAT SUPPLEMENT THE MATERIAL IN CHAPTER 12.

SUMMARY & REVIEW

12A THEORIES OF STRESS

What is the general adaptation syndrome?

The general adaptation syndrome is the predictable sequence of reactions that organisms show in response to stressors. It consists of the alarm stage, the resistance stage, and the exhaustion stage.

What are the roles of primary and secondary appraisal when people are confronted with a potentially stressful event?

Lazarus maintains that when we are confronted with a potentially stressful event, we engage in a cognitive appraisal process consisting of (1) a primary appraisal, to evaluate the

relevance of the event to our well-being (whether it will be positive, will be irrelevant, or will involve harm or loss, threat, or challenge), and (2) a secondary appraisal to determine how we will cope with the stressor.

12B SOURCES OF STRESS: THE COMMON AND THE EXTREME

How do approach–approach, avoidance–avoidance, and approach–avoidance conflicts differ?
In an approach–approach conflict, we must decide between equally desirable alternatives; in an avoidance–avoidance conflict, between two undesirable alternatives. In an approach–avoidance conflict, we are both drawn to and repelled by a choice.

How do the unpredictability of and lack of control over a stressor affect its impact?
Stressors that are unpredictable and uncontrollable are more stressful than those that are predictable and controllable.

How do people typically react to catastrophic events?
Victims of catastrophic events are initially dazed and stunned. When they begin to recover from the shock, they typically experience anxiety, nightmares, and a compulsive need to retell the event over and over.

What is post-traumatic stress disorder?
Post-traumatic stress disorder (PTSD) is a prolonged, severe stress reaction to a catastrophic event. The victim relives the trauma in flashbacks, nightmares, or intrusive memories.

12C COPING WITH STRESS

What is the difference between problem-focused and emotion-focused coping?
Problem-focused coping is a response aimed at reducing, modifying, or eliminating the source of stress; emotion- focused coping is aimed at reducing the emotional impact of the stressor.

12D EVALUATING LIFE STRESS: MAJOR LIFE CHANGES, HASSLES, AND UPLIFTS

What is the Social Readjustment Rating Scale designed to reveal?
The SRRS assesses stress in terms of life events that necessitate life change. Holmes and Rahe found a relationship between degree of life stress (as measured on the scale) and major health problems.

What roles do hassles and uplifts play in the stress of life, according to Lazarus?
According to Lazarus, daily hassles typically cause more stress than do major life changes. The positive experiences in life—the uplifts—can neutralize the effects of many of the hassles.

12E HEALTH AND DISEASE

What can cancer patients do to help themselves cope with having cancer?
Cancer patients need medical treatment and social support that will help them maintain their quality of life. They need to have opportunities to discuss their treatment and emotions openly. In addition, if patients maintain an optimistic outlook and are able to accept their condition, they may experience less distress.

What happens to a person from the time of infection with HIV to the development of full-blown AIDS?
When a person is initially infected with HIV, the body begins to produce HIV antibodies, eventually detectable in a blood test. For a period of time, the victim has no symptoms, but HIV gradually renders the immune system non-functional. The diagnosis of AIDS is made when the person succumbs to various opportunistic infections.

What are the effects of stress and depression on the immune system?
Both stress and depression have been associated with lowered immune response, and stress has been linked to increased symptoms of various infectious diseases.

What three personal factors are associated with health and resistance to stress?
Personal factors related to health and resistance to stress are optimism, psychological hardiness, and social support.

12F YOUR LIFESTYLE AND YOUR HEALTH

What constitutes an unhealthy lifestyle, and how serious a factor is lifestyle in illness and disease?
The number one health concern with most people is unhealthy lifestyle factors, which include smoking, overeating, an unhealthy diet, too much coffee or alcohol, drug abuse, and/or too little exercise and rest.

Why is smoking considered the single most preventable cause of death?
Smoking is considered the single most preventable cause of death because each year in Canada it is directly related to tens of thousands of deaths from heart disease, cancer, lung disease, and stroke.

What are some health risks of alcohol consumption?
Alcohol damages virtually every organ in the body, including the liver, stomach, skeletal muscles, heart, and brain.

What are some benefits of regular aerobic exercise?
Regular aerobic exercise reduces the risk of cardiovascular disease, increases muscle strength, moderates the effects of stress, makes bones denser and stronger, and helps maintain a desirable weight.



CHAPTER 12 • CONCEPT MAP

HEALTH AND STRESS

Health psychology examines psychological factors that contribute to health, illness, and recovery. **Stress** is the physiological and psychological response to a condition that threatens an individual and requires some form of adjustment. A **stressor** is any event capable of producing stress.

MODULE 12A THEORIES OF STRESS

Biomedical model explains illness in terms of biological factors and focuses on illness rather than health.

Selye's **general adaptation syndrome (GAS)** is a biological non-specific response to stress that consists of three stages:

- **Alarm stage:** The sympathetic nervous system mobilizes an organism to meet a threat by preparing to fight or flee.
- **Resistance stage:** There are intense physiological efforts to either resist or adapt to the stressor.
- **Exhaustion stage:** Resources become depleted and an organism may become sick or die.

Biopsychosocial model explains both health and illness in terms of a combination of biological, psychological, and social factors.

Lazarus's **cognitive theory of stress** emphasizes the importance of perception and the appraisal of stressors.

- Cognitive processes involve two types of appraisals. The physiological, emotional, and behavioural responses to an event depend on the outcome of both appraisals.
- **Primary appraisal** evaluates the meaning and importance of a potentially stressful event.
- **Secondary appraisal** evaluates one's coping resources and decides how to deal with a stressor.

MODULE 12B SOURCES OF STRESS

Conflicting Motives
Choosing between alternatives produces stress when there are conflicting motives.

- **Approach–approach** conflict arises from having to choose between equally desirable alternatives.
- **Avoidance–avoidance** conflict arises from having to choose between equally undesirable alternatives.
- **Approach–avoidance** conflict arises from having to select an alternative that has both desirable and undesirable features.

Lack of Control
People experience less stress when they think they can control a situation, even if they do not exercise that control.

Racial Stress
Members of minority groups may experience stress from being one of only a few members of a group.

Extreme Stress
Some people experience catastrophic events or are in a chronic intense situation.

Post-traumatic stress disorder (PTSD) is a prolonged and severe stress reaction characterized by anxiety, psychic numbing, withdrawal from others, and the feeling that one is reliving the traumatic experience that may follow a catastrophic event or chronic intense stress.

MODULE 12C COPING WITH STRESS

Coping refers to an individual's efforts to deal with demands that are perceived as taxing or overwhelming.

Well-functioning people use a combination of problem-focused and emotion-focused coping in almost every stressful situation.

Problem-focused coping aims to reduce, modify, or eliminate the source of stress.

Emotion-focused coping aims to reduce the emotional impact of the stressor.

MODULE 12D EVALUATING LIFE STRESS

Evaluating Major Life Events

Holmes and Rahe's Social Readjustment Rating Scale (SRRS): A person's stress level is calculated by ranking 43 life events from most to least stressful, assigning a value, and adding up the points.

Holmes and Rahe found that major life changes often precede serious illness.

Evaluating Daily Life Events

Hassles and Uplifts Scales require people to make cognitive appraisals to determine what they consider hassles (i.e., little stressors) or uplifts. Uplifts are positive experiences that can neutralize the effects of many hassles.

Lazarus believes that many hassles often produce more distress in a person's life than do major life events.

MODULE 12E HEALTH AND DISEASE

Psychoneuroimmunology studies the effects of psychological factors on the immune system.

Periods of high stress are correlated with lowered immune activity and increased symptoms of many infectious diseases.

Optimism, psychological hardiness, and social support may reduce the impact of stress on the immune system.

Optimism is an expectation that in the long run, most outcomes will be positive.

Psychological hardiness combines a sense of control over one's life, commitment to personal goals, and a tendency to view change as a challenge rather than as a threat.

Social support is people providing tangible support, information, advice, and/or emotional support in time of need.

MODULE 12F YOUR LIFESTYLE AND YOUR HEALTH

Smoking
Smoking, the most preventable cause of death in Canada, reduces the quality of life by causing chronic bronchitis, emphysema, and other respiratory diseases.

Most smokers use cigarettes to either increase their arousal or to reduce stress and anxiety.

Alcohol Consumption
Alcohol can damage almost every organ in the body, but it is especially harmful to the liver and a developing fetus.

Many people use alcohol to reduce stress and anxiety.

Aerobic Exercise
Aerobic exercise 20 to 30 minutes three or four times a week improves physical and mental fitness.

Exercise moderates the effects of stress.